

David Davidson

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Education

Niagara College

Advanced Diploma, Computer Programming & Analysis (Co-op)

Welland, ON

Jan 2023 to April 2025

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Database Systems
- **External Studies:** Fixed income markets, Rates & Credit products (Bonds, CDS, IRS, SOFR, EURIBOR), Derivative pricing (Options, Futures)

Technical Skills

Languages: Java(8, 11, 25), C++(20, 23), C#, Python, TypeScript, SQL

Frameworks: Spring Boot, Hibernate/JPA, React, ASP.NET Core, gRPC, REST, WebSocket

Developer Tools & Databases: Git/GitHub, Maven, IntelliJ, Kafka, PostgreSQL, MongoDB, JUnit, Mockito, Postman

Cloud & DevOps: AWS (EC2, S3, IAM, ELB), Docker, Kubernetes

Work Experience

Software Engineer

Eastside Creations

September 2024 to Present

Toronto, ON

- Built and maintained purchase management APIs in **Java** and **Spring Boot**, sustaining 100+ concurrent requests at sub-250ms latency.
- Profiled **JVM** performance issues in production, using heap and thread analysis to isolate contention and slow code paths.
- Refactored **Kafka Streams** to centralize transaction logs, enabling real-time audit reporting across microservices.
- Deployed backend services on **AWS EC2** and **S3**, configuring load balancers to maintain stable uptime for customer-facing APIs.
- Modelled relational schemas and tuned queries for transactional workloads, reducing redundant database round-trips and improving response times.

Software Developer

Natural by Design

January 2024 to April 2024

Welland, ON

- Implemented **Entity Framework Core** with an ORM-based data layer, managing table schemas for client and project data.
- Designed REST API endpoints in **ASP.NET Core** for account management operations, handling 200+ daily requests.
- Collaborated in Agile sprints, client presentations, and technical documentation to support software development and client delivery.

Projects

Bond Pricing Engine | Java 25, Concurrency, GC Tuning, gRPC, Fixed Income

- Built a multithreaded pricing engine calculating dirty and clean prices, accrued interest, and YTM using a **Newton-Raphson solver**.
- Derived **Macaulay duration**, **modified duration**, and **DV01** from the pricing, reusing cashflows and discount factors so each bond is valued once.
- Configured **ZGC** heap sizing and GC logging to minimize pause times, achieving sub-50ms latency across bulk pricing runs.
- Architected a **gRPC** layer exposing pricing and sensitivity analytics through unary and server-streaming RPCs with **Protocol Buffers**.

Order Management System | Java 25, LMAX Disruptor, Netty, FIX, Low-Latency Architecture

- Built a **price-time priority matching engine** over **TreeMap**-backed order books, supporting partial fills and O(1) cancels.
- Designed an event-driven pipeline on **LMAX Disruptor** ring buffers with a **zero-allocation hot path**, benchmarked at **8.2M orders/sec** via **JMH**.
- Engineered a hand-rolled **FIX** gateway parsing tag-value pairs (NewOrderSingle, OrderCancelRequest) and publishing orders onto the inbound ring buffer.
- Streamed live market data over a **Netty WebSocket** server to a **React** UI with a real-time depth ladder and trade tape.
- Building a pluggable strategy interface for **algorithmic trading agents** to connect to the book and submit orders programmatically, modeled on broker-style agent integration.

Document Processing Software | Java 21, Spring Boot, React, PostgreSQL, Python, Docker

- Engineered a full-stack application processing 50+ legal documents with cloud storage, reducing review time by ~75%.
- Developed a **Python** NLP microservice using **GPT-4** to extract entities and generate summaries from legal documents.
- Created a **React** frontend with **Vite** and **Tailwind** to display AI-generated summaries, document data and extracted fields.